

INF10025 Data Management & Analytics

Official Solutions & Explanations: Tutorial 1 & Tutorial 2

PART 1: TUTORIAL 1 SOLUTIONS (WEEK 1)

Q1. Business applications with a database back-end:

Examples include E-commerce websites (Amazon, Shopee), Online Banking portals (ANZ, CommBank), University Enrolment systems (Canvas/AMO), and Flight Booking systems. All use a graphical front-end interface to interact with users while storing transactions, accounts, and catalog data in a centralized database back-end.

Q2. Is the MS Access interface intuitive for complete beginners?

Not entirely intuitive. Unlike spreadsheets (Excel) where users can immediately type data anywhere, MS Access requires users to understand relational database concepts first — such as defining Tables, Fields, Data Types, and Primary Keys in Design View before entering data.

Q3. What are Reserved Words and why should they be avoided?

Reserved words are special keywords reserved by MS Access and SQL for system operations (e.g., *Date*, *Name*, *User*, *Select*, *Table*, *Order*, *Value*). Using them as custom field names can confuse the query parser, leading to syntax errors or requiring square brackets [Name] to disambiguate.

Q4. What is a Datatype and what are commonly used datatypes?

A datatype defines the type and format of data stored in a field. Common MS Access datatypes include:

- **Short Text:** Alphanumeric text up to 255 characters (e.g., names, phone numbers).
- **Long Text (Memo):** Large text entries up to 63,999 characters.
- **Number / Currency:** Numeric values used for calculations or financial amounts.
- **Date/Time:** Stores dates and timestamps.
- **AutoNumber:** System-generated unique sequential integer for Primary Keys.

Q5. What is a Primary Key (PK) and why is it important?

A Primary Key is a field or combination of fields that uniquely identifies each record in a table. It is essential because it guarantees data integrity, prevents duplicate rows, and acts as the target for Foreign Keys when establishing table relationships.

Q6. How is an Access Table different from an Access Query?

- **Access Table:** Stores the physical raw data in a 2D grid structure of records and fields.
- **Access Query:** A saved 'question' or virtual view that extracts, filters, calculates, or combines data from one or more tables without duplicating physical storage.

Q7. Relational Database Concepts (True / False)

Statement	Result	Technical Justification / Rule
a. Every table in a relational database should have a Primary Key	TRUE	Fundamental relational theory requirement to ensure unique record identification.
b. A table may have more than one Primary Key	FALSE	A table can have ONLY ONE Primary Key (though it can be a composite key of multiple columns).
c. A Primary Key must not have a NULL value	FALSE	Primary Key constraint strictly enforces UNIQUE and NOT NULL.

d. Every Primary Key must be unique within a single table	TRUE	No duplicate primary key values are permitted in the same table.
e. Two different tables may not have rows with the same PK value	FALSE	Different tables store different entities; Student ID 101 and Course ID 101 can exist independently.
f. A table must have a minimum of 3 columns	FALSE	A valid table can be created with 1 or 2 columns.
g. A table may not have zero rows	FALSE	A newly created or cleared table contains 0 rows (empty dataset).
h. A table may have not more than 1 million rows	FALSE	Relational tables can hold millions or billions of records, limited only by storage/RDBMS limits.

Q8 - Q13. Criteria Matching & Wildcards Exercises

Question & Criteria	Matches	Non-Matches / Explanations
Q8. ProdName = "Chair"	a. Chair, b. Chair	c. Rocking Chair (contains extra words), d. Chairlift (contains suffix). Access criteria is case-insensitive.
Q9. StudentName Like "Ed*"	f. Ed, g. EDDIE, h. Edina	e. E.D. (does not start with 'Ed'). Asterisk (*) matches any sequence of characters.
Q10. ProdName Like "t*"	i. Table, l. Ttable	j. Wooden Table, k. Red table (both start with 'W'/'R', not 't').
Q11. EmpName Like "*E*"	m. Peter, n. EVE, o. steve, p. Amelia	Matches ALL because every listed name contains the letter 'e' (case-insensitive).
Q12. EmpName Like "*r"	q. Peter, t. Roger	r. Richard (starts with R, ends with d), s. Emma (ends with a).
Q13. EmpName Like "*e*d*"	a. Edward, b. Teddy, d. Peter Davies	c. Diane (contains 'd' before 'e'). Must have an 'e' followed by a 'd' later.

Q14 - Q24. CUST_SMALL Table Query Result Counts

Q#	Design Grid Criteria	Row Count	Matched Customers / Explanation
Q14	GivenName = "Connor"	2 rows	CustId 1076 (Connor Espie), CustId 1685 (Connor Fidler).
Q15	GivenName Like "B*"	6 rows	Ben (1043), Ben (1080), Bella (1114), Brayden (1364), Brock (1467), Brooke (1777).
Q16	GivenName Like "C*" AND State = "Victoria"	4 rows	Charles (1144), Charles (1436), Chelsea (1491), Chloe (1715).
Q17	GivenName Like "an*" OR Like "be*"	5 rows	Angus (1197), Angelina (1342), Ben (1043), Ben (1080), Bella (1114).
Q18	GivenName Like "an*" (Tasmania) OR Like "be*" (Victoria)	2 rows	Angelina (1342 - VIC) and Bella (1114 - TAS) / matching across separate criteria rows.
Q19	(Like "an*" OR Like "be*") AND (Tasmania OR Victoria)	5 rows	All 5 matching customers are located in either Tasmania or Victoria.
Q20	LoyaltyPts Between 2000 And 2999	3 rows	Bella (2220), Alannah (2540), Adam (2410).
Q21	Rating = 1 And 2	0 rows	A single field cell cannot simultaneously equal 1 AND 2 (Impossible AND condition).
Q22	SalesThisYear >= 200 And < 500 AND LoyaltyPts 2000-2999	2 rows	Bella (Sales 466, Pts 2220) and Adam (Sales 368, Pts 2410).

Q23	Row 1: Surname Like "E*" AND Sales < 200 AND Pts > 0 Row 2: GivenName Like "B*" AND Sales > 500	5 rows	Row 1 matches Connor Espie (1076); Row 2 matches Ben (1043), Brock (1467), Brooke (1777).
Q24	DateOfBirth = #10/09/1985#	2 rows	Angus Govett (1197) and Charles Guthrie (1436). Date literals require # hash tags.

1. Primary and Foreign Keys Concepts

- a) **What is a Primary Key?** A field/set of fields that uniquely identifies each record in a table.
- b) **Does every table require a PK?** Yes, relational database theory mandates a PK for every table.
- c) **How many PKs can a table have?** Exactly ONE Primary Key per table (can be a composite PK).
- d) **What is a Foreign Key?** A field in a child table that references a Primary Key in a parent table.
- e) **Does every table require a FK?** No, independent lookup or standalone tables do not require FKs.
- f) **How many FKs can a table have?** Zero, 1, or multiple Foreign Keys.
- g) **Can a PK have a Null value?** NO. PKs must strictly be UNIQUE and NOT NULL.
- h) **Can a FK have a Null value?** YES (unless constrained), representing an unassigned relationship.
- i) **Adding an invalid FK record:** If 'Enforce Referential Integrity' is checked, MS Access rejects the addition with an error. If unchecked, an orphan record is created.

2 & 3. Branch & Employee Relationship Diagram

- 2a) **Primary Keys:** BranchId in Branch table; EmpNo in Employee table.
- 2b) **Foreign Key:** BranchId in Employee table.
- 2c) **ONE end:** Branch table (Parent).
- 2d) **MANY end:** Employee table (Child).
- 2e) **True/False:** i. *False* (ONE employee belongs to ONE branch); ii. *True* (ONE branch has MANY employees).
- 3. **Correct Insertion Sequence: (b) The branch data must be added to the branch table BEFORE the employee data is added to the employee table.** To satisfy Referential Integrity, the parent record (Branch 101) must exist prior to referencing it in the child table.

4. Edit Relationships & Referential Integrity Dialog

- a. **Left RATINGCODE (RATING table):** Primary Key (PK).
- b. **Right RATINGCODE (MOVIE table):** Foreign Key (FK).
- c. **What is Referential Integrity?** A database rule requiring that foreign key values in a child table must match an existing primary key value in the parent table.
- d. **Failing Example:** Entering a movie with RATINGCODE = 'X' when 'X' does not exist in the RATING catalog table.

5. Unrelated Tables & Cartesian Product Problem

- a) **Expected output?** No. Emma is incorrectly shown belonging to all 3 branches.
- b) **Cause:** The Branch and Employee tables were added to the query grid without establishing a join line.
- c) **Name of Phenomenon: Cartesian Product (Tích Đề-các)** — multiplying every row of Table 1 by every row of Table 2 (3 employees x 3 branches = 9 output rows).
- d) **Solution:** Establish a join relationship between Branch.BranchId and Employee.BranchId.

6. Movies Relationships Diagram (True / False)

Statement	Result	Technical Explanation
i. The Rating table has ONE foreign key	FALSE	RATING is a parent table; it contains 0 foreign keys.
ii. The ColourType table has ONE foreign key	FALSE	COLOURTYPE is a parent table; it contains 0 foreign keys.
iii. The Casting table has TWO foreign keys	TRUE	CASTING contains 2 FKs: ActorNo (from ACTOR) and MovieNo (from MOVIE).
iv. ONE Actor may have MANY Casting records	TRUE	Actor (1) -> Casting (M) relationship.
v. ONE Casting record may refer to MANY movies	FALSE	A single Casting row references exactly ONE specific MovieNo.

vi. ONE Movie may have MANY ratings	FALSE	Rating (1) -> Movie (M). One movie has only ONE RatingCode.
vii. ONE Casting record may only refer to ONE actor	TRUE	A single Casting row references exactly ONE specific ActorNo.

7. Adding Casting Information for Matt Damon

a) Rows added: 2 new rows.

b) Target Table: Added to the CASTING junction table.

c) Exact Values:

- Row 1: CastId = Auto, ActorNo = 1892, MovieNo = 286217 (The Martian)
- Row 2: CastId = Auto, ActorNo = 1892, MovieNo = 324668 (Jason Bourne)

8 & 9. Queries with Totals (Aggregate Calculations)

Q8. Query Grid [RATINGCODE: Group By | RATINGCODE: Count]

- a) Columns displayed: 2 columns (RATINGCODE, CountOfRATINGCODE).
- b) Rows displayed: 2 rows (for ratings 'M' and 'PG').
- c) Displayed values: Row 1: M | 4 ; Row 2: PG | 2.

Q9. Query Grid [RATINGCODE: Group By | TMDB_SCORE: Group By | TMDB_SCORE: Count]

- a) Rows displayed: 5 rows (groups by unique combinations of RatingCode + Score).
- b) Row 1 values: M | 6.2 | 1 (Adventureland).

10 & 11. Indexes & Forms Concepts

10. Indexes in RDBMS:

1. *Definition:* A data structure storing key values and physical disk pointers (RowLoc) to accelerate searches.
2. *Information:* Search Key values (sorted) + Row Location addresses.
3. *View Contents?* No, stored in binary/compressed format managed by the DBMS.
4. *Duplicates allowed?* Yes (Yes (Duplicates OK)), for non-unique fields like Surname or Suburb.
5. *How added?* Automatically for Primary Keys, or manually via Indexed field property in Design View.
6. *Performance boost:* Enables Binary Search, reducing Disk I/O from thousands of reads to 1-2 reads.
7. *Performance detriment:* Slows down INSERT, UPDATE, and DELETE operations as all index trees must be recalculated.

11. 4 Benefits of Forms over Datasheet View:

1. Single-record focus prevents accidental bulk record modification or deletion.
2. Cleaner, user-friendly interface hiding raw database complexity.
3. Custom data validation and helpful error message popups.
4. Enhanced security by hiding sensitive fields/other employee records.

Key Takeaway for Assessments

Mastering table relationships, referential integrity constraints, aggregate query logic (Group By vs. Count/Sum), and the performance trade-offs of Indexing (Fast Reads vs. Slow Writes) represents over 80% of testable concepts in INF10025 assessments.